B E N C H M A R K R E P O R T

They All Saw the Grit

But Only 15 Traced It Upstream

Behavioral Analysis of 22 LLMs on an Upstream Root Cause Reasoning Task

22	68.2%	7/7	2
LLMs Tested	Pass Rate	Saw Grit, Dismissed It	Models Pass All 4

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Kaggle Benchmarks | Ceyhan WWTP, Adana, Turkey | March 2026

Part of the WWTP Benchmark Series | Dewatering System Root Cause Module

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1. Executive Summary

I posed a dewatering system fault diagnosis question to 22 LLMs: given six symptoms of a failing centrifugal decanter, identify the upstream root cause from four options. The correct answer is **C) Grit removal system not working properly.**

The pass rate is 68.2% (15/22). This is the lowest pass rate in the series after the Biogas benchmark (14.3%), and it reveals a specific reasoning limitation: the inability to trace symptoms upstream through the process chain.

F1 — They All Saw the Grit: All 7 models that chose the wrong answer (A — bearing failure) mentioned grit in their responses. Every single one considered and then dismissed it. They recognized the correct answer but did not follow the upstream logic that connects grit ingress to the observed symptoms.

F2 — Local vs. Upstream Reasoning: The “A trap” is fundamentally different from the traps in previous benchmarks. Models that chose A treated symptoms as direct evidence of a local failure (bearings cause vibration, bearings cause scratches). Models that chose C traced the causal chain upstream (grit enters → scratches blades → causes imbalance → causes vibration). The question asks for ROOT CAUSE — upstream reasoning was required.

F3 — R1’s Think Block Length Predicts Outcome: In the previous Pump RCA benchmark, DeepSeek R1 wrote 21,147 characters of internal reasoning and found the correct answer. Here, R1 wrote only 2,171 characters — 9.7x shorter — and chose wrong. On this evidence, R1’s shorter deliberation appeared to correlate with a worse outcome, though more data would be needed to confirm this pattern.

F4 — DeepSeek V3.2 Breaks Its Streak: V3.2 had passed all 3 previous benchmarks (Material Selection, Pump RCA, Biogas Desulfurization). Here it failed, choosing A. This is a reminder that past performance on different questions does not guarantee future results.

F5 — Only 2 Models Pass All 4 Benchmarks: Across the full WWTP series, only Qwen3 235B A22B Instruct and Qwen3 Next 80B Instruct have passed every question. Neither is a flagship model. No Claude, no Gemini Pro, no DeepSeek R1.

2. The Question

This benchmark tests a specific cognitive skill: upstream root cause tracing. The scenario presents a centrifugal decanter (sludge dewatering unit) with six symptoms. The correct answer requires understanding that the visible damage to the decanter is not caused by the decanter itself, but by a failure in the upstream grit removal system.

Table 1: The Causal Chain (What the Correct Answer Requires)

Step	Event	Observable Symptom
1	**Grit pumps fail → grit is not removed from sludge**	*(Not directly visible)*
2	Grit-laden sludge enters the centrifugal decanter	*(Not directly visible)*
3	Abrasive grit scratches the scroll conveyor blades	Symptom 4: visible scratches and wear marks
4	Worn blades create uneven mass distribution	Symptom 5: unbalanced rotation
5	Imbalance causes mechanical vibration	Symptom 1: excessive vibration
6	Damaged scroll cannot convey solids effectively	Symptoms 2, 3, 6: low efficiency, high polymer, wet cake

Table 1: The correct reasoning requires 6 steps of upstream logic. Models that chose A stopped at step 3–5, treating the symptoms as the cause rather than tracing back to step 1.

Table 2: Answer Options

	Cause	Reasoning Type	Assessment
A	Scroll bearings need replacement	Local — treats symptoms as cause	THE TRAP — explains symptoms but not their origin
B	Polymer dosing malfunction	Chemical — partial match	Would not cause mechanical wear or vibration
C	**Grit removal system failure**	**Upstream — traces to root cause**	**CORRECT — explains all 6 symptoms via causal chain**
D	Digester mixing problem	Upstream but wrong domain	Would affect sludge quality, not cause abrasion

3. Results

Table 3: Answer Distribution

Answer	Count	Status	Models
C	15	CORRECT	All Claude (5), All Qwen (4), Gemini 2.5 Pro, Gemini 3 Pro/Flash, Gemma 1B/12B, DeepSeek V3.1
A	7	WRONG — the “A” trap	DeepSeek R1, DeepSeek V3.2, Gemini Flash/Flash Lite/2.5 Flash, Gemma 4B/27B

Table 3: No model chose B or D. The competition was entirely between A (local) and C (upstream). DeepSeek R1 and Gemma 27B also have parsing issues but their visible answers are A regardless.

Figure 1: Answer Distribution — C (Correct) vs. A (Trap)

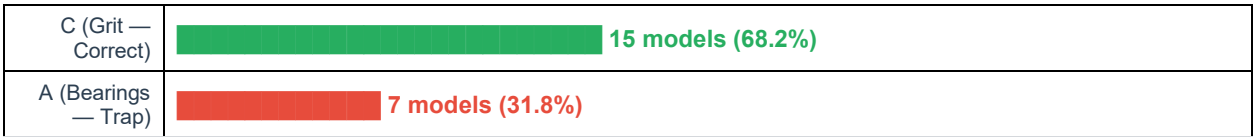

Figure 1: A clean two-way split. No model chose B (polymer) or D (digester). The entire competition was between upstream reasoning (C) and local reasoning (A).

Table 4: Complete Results — All 22 Models

Model	Ans	Status	Out Tok	Latency	Terms	Family
Claude Haiku 4.5	C	PASS	386	4.9s	10	Claude
Claude Opus 4.1	C	PASS	368	13.3s	10	Claude
Claude Opus 4.5	C	PASS	390	10.1s	8	Claude
Claude Sonnet 4	C	PASS	373	9.2s	6	Claude
Claude Sonnet 4.5	C	PASS	402	10.6s	8	Claude
DeepSeek V3.1	C	PASS	469	2.7s	9	DeepSeek
Gemini 2.5 Pro	C	PASS	402	24.5s	8	Gemini
Gemini 3 Flash Preview	C	PASS	324	9.3s	9	Gemini
Gemini 3 Pro Preview	C	PASS	335	23.5s	9	Gemini
Gemma 3 1B*	C	PASS	1	1.1s	0	Gemma
Gemma 3 12B	C	PASS	210	11.0s	7	Gemma
Qwen3 235B A22B Instruct	C	PASS	246	3.2s	7	Qwen
Qwen3 Coder 480B	C	PASS	271	2.9s	7	Qwen
Qwen3 Next 80B Instruct	C	PASS	254	2.3s	9	Qwen
Qwen3 Next 80B Thinking	C	PASS	1,774	9.9s	7	Qwen
DeepSeek R1 (CoT)	A	FAIL	930	6.5s	5	DeepSeek
DeepSeek V3.2	A	FAIL	154	7.6s	5	DeepSeek
Gemini 2.0 Flash	A	FAIL	322	3.2s	5	Gemini
Gemini 2.0 Flash Lite	A	FAIL	120	2.0s	3	Gemini

Gemini 2.5 Flash	**A**	**FAIL**	386	10.4s	5	Gemini
Gemma 3 4B	**A**	**FAIL**	320	8.3s	5	Gemma
Gemma 3 27B	**A***	**FAIL**	380	10.8s	6	Gemma

Table 4: All 22 models. A for Gemma 27B indicates markdown formatting (**A**) caused a parsing mismatch, but the answer is A regardless. *Gemma 3 1B passed with a single token "C" and zero explanation — correct answer but no evidence of reasoning.*

4. LLM Behavior Analysis

4.1 The "A" Trap — Local Diagnosis vs. Upstream Reasoning

Seven models independently chose A (bearing failure) with remarkably similar reasoning. The most striking observation is not that they missed the correct answer — it is that they all considered it and rejected it.

Table 5: The Common Reasoning Chain in A-Choosing Models

Step	Reasoning	Error Analysis
1	Vibration + unbalanced rotation → mechanical failure	Correct observation, but stops at local level
2	Scratches on blades → bearing misalignment causes scraping	**THE FORK — treats scratches as effect, not evidence of cause**
3	*Consider grit (C) — "could contribute to wear over time"*	Acknowledged but dismissed — 7 out of 7 models
4	Dismiss C: "grit wouldn't directly cause vibration" or "secondary"	**Critical error — grit DOES cause vibration via wear chain**
5	Conclude: bearings are "the most cohesive explanation"	Locally cohesive, but not root cause

The critical fork is at Step 2. When models encounter "scratches on scroll blades," they face a choice: interpret the scratches as caused by bearing misalignment (local, looking down), or interpret them as caused by abrasive material entering the system (upstream, looking back). The 15 correct models asked "what could scratch hard steel?" and arrived at grit. The 7 incorrect models asked "what else vibrates?" and arrived at bearings.

A pattern observed across the WWTP series: this is the third benchmark where models follow a reasoning shortcut that appears logical in isolation but misses the operational context. In Biogas: "SOB = aerobic → aeration basin." In Pump RCA: "high current = electrical." Here: "vibration = bearings." Each trap shares a common structure: the model anchors on the most salient symptom and reasons forward rather than backward through the process chain.

4.2 R1's Think Block — When Shorter Reasoning Correlates with Worse Outcomes

DeepSeek R1's behavior on this benchmark provides an interesting comparison point with the previous Pump RCA benchmark. The contrast is striking:

Table 6: R1 Think Block Comparison Across Two Benchmarks

Dimension	Pump RCA (Previous)	Dewatering RCA (This)
Think block length	21,147 chars (85 paragraphs)	2,171 chars (7 paragraphs)
Deliberation ratio	1x (baseline)	**9.7x shorter**
Considered correct answer?	Yes — 4 times during deliberation	Yes — mentioned grit once, dismissed it
Changed mind?	Yes: A → C → A (found way back)	No — committed to A early
Final visible answer	**A (correct)**	**A (wrong)**
Outcome	Extended reasoning rescued correct answer	Short reasoning locked in wrong answer

> *Grit (cause C) could wear components and reduce efficiency, but it wouldn't typically cause unbalanced rotation unless it's unevenly distributed.*
>
> — DeepSeek R1, <think> block on Dewatering RCA — dismissing the correct answer in one sentence

Figure 2: R1 Think Block Length — Two Benchmarks, Two Outcomes

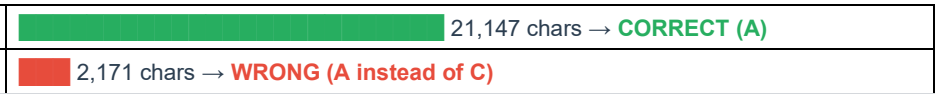

Figure 2: R1's think block was 9.7x shorter on the question it got wrong. On the Pump RCA, extended deliberation allowed R1 to traverse a wrong answer and return to the right one. Here, shorter reasoning appeared to correlate with premature commitment.

This is a single comparison across two questions and should not be over-interpreted. However, the pattern is suggestive: on the Pump RCA question, R1's extensive deliberation (21K characters) allowed it to traverse a wrong answer and return to the right one. Here, with only 2K characters of deliberation, R1 committed to the wrong answer early and never revisited it. If this pattern holds across more benchmarks, it could suggest that CoT reasoning's value depends partly on the amount of computation allocated to the task.

4.3 The V3.2 Streak Breaks — Past Performance Does Not Guarantee Future Results

DeepSeek V3.2 had achieved something no other model managed: passing all three previous WWTP benchmarks (Material Selection, Pump RCA, and Biogas Desulfurization). On this fourth question, it failed — choosing A (bearings) with a concise but incorrect explanation.

V3.2's response is notably short (154 tokens) and commits to A without extended deliberation. It mentions grit as a possibility but dismisses it as a "secondary" contributor. This

contrasts with V3.2's behavior on the Biogas benchmark, where it was one of only 3 models to answer correctly — suggesting that its domain strengths may be unevenly distributed across topics.

This is an important reminder for the series: a model that passes three questions may fail the fourth. Consistency across domain tasks is rare, and the only way to discover it is to test broadly.

4.4 Family Behavioral Patterns

Table 7: Family-Level Results

Family	Pass	Rate	Avg Out	Avg Terms	Observation
Claude	5/5	**100%**	384	8.4	Consistent upstream reasoning across all sizes
Qwen	4/4	**100%**	636*	7.5	Thinking model inflates average (*excl: 257)
Gemini	3/6	**50%**	315	6.5	Pro models pass, Flash models fail — tier split
Gemma	2/4	**50%**	228	4.5	Inverted scaling: 1B/12B pass, 4B/27B fail
DeepSeek	1/3	**33%**	518	6.3	V3.1 pass, V3.2 and R1 fail

Figure 3: Family Pass Rates on This Benchmark

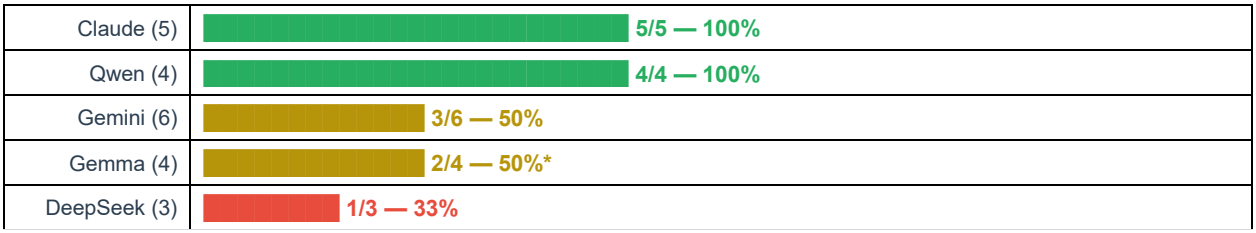

*Figure 3: Family pass rates. Claude and Qwen achieved 100%. *Gemma's 50% includes Gemma 1B which passed with a single token "C" and zero explanation.*

from the previous benchmarks. Both families achieved 100% on this question. Claude's consistency is particularly notable: all five models used upstream reasoning, mentioned abrasive grit, and linked it to the full symptom chain. This family-level consistency has now held across 4 benchmarks (with the exception of Biogas, where Claude's knowledge gap was domain-specific).

Gemini shows a clear tier split: the newer/larger models (2.5 Pro, 3 Flash Preview, 3 Pro Preview) all passed, while the older/smaller models (2.0 Flash, 2.0 Flash Lite, 2.5 Flash) all failed. This suggests that upstream reasoning capability may have improved across model generations in the Gemini family, though the sample is small.

Gemma's inverted scaling is puzzling: the 1B model passed (with a single token "C" and zero explanation), the 12B model passed with reasoning, the 4B model failed, and the 27B model failed. This does not follow the expected size-performance gradient and may indicate that

Gemma 1B's correct answer is coincidental rather than knowledge-based. The 27B failure is particularly unexpected given its success on the Pump RCA benchmark.

A note on sample sizes: *family-level observations are based on unequal representation (Claude 5, Gemini 6, Qwen 4, Gemma 4, DeepSeek 3). These patterns should be treated as preliminary observations.*

4.5 Thinking Models — Mixed Signals on This Task

Table 8: Thinking vs. Standard Model Comparison

Model	Out Tok	Latency	Answer	Status	Type
DeepSeek V3.1	469	2.7s	C	PASS	Standard
DeepSeek R1 (CoT)	930	6.5s	A	FAIL	Thinking
Qwen3 Next 80B Instruct	254	2.3s	C	PASS	Standard
Qwen3 Next 80B Thinking	1,774	9.9s	C	PASS	Thinking

The two thinking models diverge on this question. Qwen3 Thinking answered correctly (C) but used 7x more tokens than its Instruct counterpart (1,774 vs 254). DeepSeek R1 answered incorrectly (A) with 2x more tokens than V3.1 (930 vs 469). In this benchmark, thinking helped one model (Qwen) and did not help the other (R1). The pattern from the Biogas benchmark (both thinking models failed) and the Pump RCA benchmark (R1's extended reasoning saved it) continues to resist simple generalization.

5. Cross-Benchmark Comparison — Four Questions, One Model Set

Table 9: Four WWTP Benchmarks Side by Side

Dimension	Material Sel.	Pump RCA	Dewatering RCA	Biogas Desulf.
Pass Rate	**100%**	77–82%	**68.2%**	14.3%
Trap Type	None	"C" electrical	**"A" local bearing**	"D" aerobic
Skill Tested	Basic knowledge	Symptom elimination	**Upstream reasoning**	Domain depth
Claude	5/5	5/5	5/5	0/4
Qwen	4/4	4/4	4/4	2/4
DeepSeek	3/3	2–3/3	1/3	1/3
R1 (CoT)	Pass, high cost	Correct, parsed wrong	**Genuinely wrong**	Failed

Table 10: Models That Passed All 4 Benchmarks

Model	Material	Pump RCA	Dewatering	Biogas

Qwen3 235B A22B Instruct	PASS	PASS	PASS	PASS
Qwen3 Next 80B Instruct	PASS	PASS	PASS	PASS

Table 10: Only 2 of 22 models passed all 4 WWTP benchmarks. Both are mid-tier Qwen Instruct models. No flagship model from any family achieved this.

Figure 4: The Narrowing Funnel — Pass Rates Across 4 WWTP Benchmarks

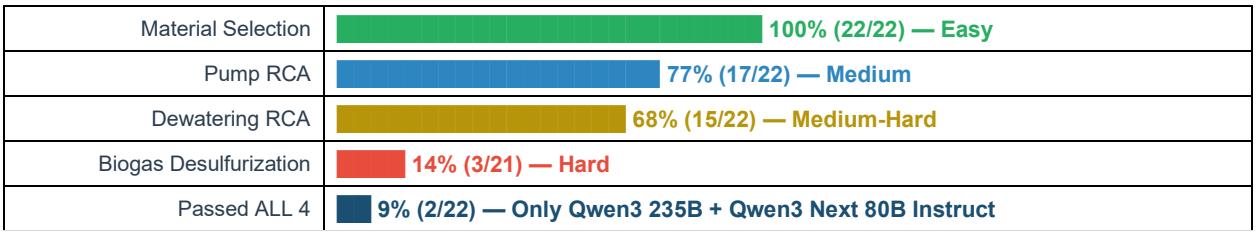

Figure 4: The difficulty gradient reveals different layers of capability. Easy questions separate no one. Hard questions separate nearly everyone. The full series separates all but 2.

The narrowing funnel is dramatic: 22 models passed the easy question, ~17 passed the medium ones, and only 2 have passed everything. This reinforces the series' central finding: general capability rankings do not predict domain performance. The two survivors are not the largest, most expensive, or highest-ranked models — they are mid-tier instruct-tuned models that happen to combine the right reasoning patterns for these specific tasks.

6. Conclusions

This fourth WWTP benchmark adds a new dimension to the series: upstream reasoning. Where the Pump RCA benchmark tested symptom elimination and the Biogas benchmark tested domain depth, this question specifically tested whether models can trace a causal chain backward through a process system.

The "A trap" reveals a reasoning pattern that may be worth investigating further: when models encounter symptoms that fit a local cause, they may anchor on that explanation without asking whether a more upstream cause could produce the same symptoms through a longer chain. All 7 models that chose A explicitly mentioned grit but dismissed it. They had the information but did not follow it upstream.

R1's contrasting behavior across two benchmarks (21K chars → correct vs. 2K chars → wrong) is a data point in a growing conversation about CoT reasoning. It does not prove that more tokens produce better answers, but it is consistent with a hypothesis that insufficient deliberation on complex questions may lead to premature commitment to a wrong answer.

The most practically useful finding for model selection: only two models have now survived all four WWTP questions. Both are Qwen Instruct models. This is one observation from one benchmark series, but it aligns with a broader pattern in these reports: the models that perform best on domain tasks are not always the ones that top general leaderboards.

7. Methodology Notes

Non-determinism: LLMs produce different outputs on repeated runs. This test used a single run per model. Results should be interpreted as indicative rather than absolute. Future iterations of this series may incorporate multi-run protocols (3–5 runs per model) to assess result stability and provide confidence intervals.

Single question: This report analyzes one question from a broader benchmark suite. Family-level and reasoning-pattern conclusions should be validated across the full question set.

API conditions: All models were accessed through the Kaggle Benchmarks platform API with default temperature settings. Different configurations may produce different results.

Parsing notes: DeepSeek R1's response begins with a <think> block (first char "<"). Gemma 27B's response begins with bold markdown (**A**, first char "*"). Both models' visible answers are A — adjusting for parsing would not change their pass/fail status.

Cross-benchmark comparison: The 4-benchmark matrix references results from independently published reports. The Biogas benchmark used 21 models (Gemini 2.5 Pro was not included). All others used 22 models.